```
!git clone https://github.com/trextrader/hotdogornot
```

```
Cloning into 'hotdogornot'...
remote: Enumerating objects: 16300, done.
remote: Total 16300 (delta 0), reused 0 (delta 0), pack-reused 16300 (from 3)
Receiving objects: 100% (16300/16300), 553.38 MiB | 47.06 MiB/s, done.
Resolving deltas: 100% (2224/2224), done.
Updating files: 100% (14491/14491), done.
```

```
cd hotdogornot/
```

```
/content/hotdogornot
```

```
cd training
```

```
/content/hotdogornot/training
```

```
# @title
!pip install -e ".[dev]"
```

```
Downloading icrawler-0.6.10-py3-none-any.whl (36 kB)
Downloading ImageHash-4.3.2-py2.py3-none-any.whl (296 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 296.7/296.7 kB 19.7 MB/s eta 0:00:00
Downloading onnx-1.21.0-cp312-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (17.6 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 17.6/17.6 MB 78.1 MB/s eta 0:00:00
Downloading onnxruntime-1.26.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (18.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 18.2/18.2 MB 79.5 MB/s eta 0:00:00
Downloading pytest_cov-7.1.0-py3-none-any.whl (22 kB)
Downloading streamlit-1.57.0-py3-none-any.whl (9.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 9.2/9.2 MB 85.3 MB/s eta 0:00:00
Downloading trimesh-4.12.2-py3-none-any.whl (741 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 741.0/741.0 kB 47.5 MB/s eta 0:00:00
Downloading ultralytics-8.4.48-py3-none-any.whl (1.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.2/1.2 MB 58.1 MB/s eta 0:00:00
Downloading coverage-7.14.0-cp312-cp312-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (254 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 254.6/254.6 kB 26.7 MB/s eta 0:00:00
Downloading primp-1.2.3-cp310-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (4.5 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 4.5/4.5 MB 89.3 MB/s eta 0:00:00
Downloading pydeck-0.9.2-py2.py3-none-any.whl (11.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 11.3/11.3 MB 97.8 MB/s eta 0:00:00
Downloading ultralytics_thop-2.0.19-py3-none-any.whl (28 kB)
Downloading bs4-0.0.2-py2.py3-none-any.whl (1.2 kB)
Building wheels for collected packages: rfconnectorai
  Building editable for rfconnectorai (pyproject.toml) ... done
  Created wheel for rfconnectorai: filename=rfconnectorai-0.1.0-py3-none-any.whl size=1419 sha256=734dd19f12e07e19ae1d25f237084bec5d574d8c6d3
  Stored in directory: /tmp/pip-ephem-wheel-cache-eruao6cm/wheels/3b/ac/cd/b6bca6a547c755dbdfd5b88fac5a137fb63d02b6acb225621a
Successfully built rfconnectorai
Installing collected packages: trimesh, primp, onnxruntime, coverage, pydeck, onnx, imagehash, ddgs, bs4, pytest-cov, icrawler, ultralytics-t
Successfully installed bs4-0.0.2 coverage-7.14.0 ddgs-9.14.2 icrawler-0.6.10 imagehash-4.3.2 onnx-1.21.0 onnxruntime-1.26.0 primp-1.2.3 pydec
```

```
!git pull
```

```
Already up to date.
```

```
%cd /content/hotdogornot
!pwd
!ls training/data/labeled/embedder | head
```

```
/content/hotdogornot
/content/hotdogornot
2.4mm-F
2.4mm-M
2.92mm-F
2.92mm-M
3.5mm-F
3.5mm-M
```

```
%%shell
  set -euxo pipefail
  cd /content/hotdogornot
```

```
export PYTHONPATH=/content/hotdogornot/training
export PYTHONUNBUFFERED=1

python -u -m rfconnectorai.data.audit \
    --data-dir training --out docs/DATASET_AUDIT.md

echo "----- audit summary -----"
head -40 docs/DATASET_AUDIT.md

python -u -m rfconnectorai.data.crop_instances \
    --input training/data/labeled/embedder \
    --manifest datasets/rfconnectors/instances.jsonl \
    --out datasets/rfconnectors/crops \
    --mode whole-image --base-dir training

wc -l datasets/rfconnectors/instances.jsonl
```

```
+ cd /content/hotdogornot
+ export PYTHONPATH=/content/hotdogornot/training
+ PYTHONPATH=/content/hotdogornot/training
+ export PYTHONUNBUFFERED=1
+ PYTHONUNBUFFERED=1
+ python -u -m rfconnectorai.data.audit --data-dir training --out docs/DATASET_AUDIT.md
audit report: docs/DATASET_AUDIT.md
audit json:   docs/DATASET_AUDIT.json
+ echo '----- audit summary -----'
----- audit summary -----
+ head -40 docs/DATASET_AUDIT.md
# Dataset Audit

- Generated: `2026-05-12T06:59:49.996159+00:00`
- Data dir: `/content/hotdogornot/training`
- Roots audited: 5

## Summary By Root

| Root | Images | Videos | Synthetic | Holdout | Reference | Multi-conn hint | Unreadable |
|---|---:|---:|---:|---:|---:|---:|---:|
| `/content/hotdogornot/training/Images` | 55 | 0 | 0 | 0 | 0 | 1 | 0 |
| `/content/hotdogornot/training/data/labeled` | 13850 | 0 | 0 | 0 | 0 | 0 | 0 |
| `/content/hotdogornot/training/data/test_holdout` | 8 | 0 | 0 | 8 | 0 | 0 | 0 |
| `/content/hotdogornot/training/data/reference` | 2 | 0 | 0 | 0 | 2 | 0 | 0 |
| `/content/hotdogornot/training/data/videos` | 0 | 3 | 0 | 0 | 0 | 0 | 0 |

## Duplicate Groups (sha256)

- `Images/Sb13fdff1dd1f4fc5841bee47aee102f4F (1).jpg`, `Images/Sb13fdff1dd1f4fc5841bee47aee102f4F.jpg`
- `Images/Sc0620eb6b3484352ba605af77b62f28cu (1).jpg`, `Images/Sc0620eb6b3484352ba605af77b62f28cu.jpg`
- `Images/20-Type SMA Adapter Kit – Universal SMA to UHF SO-239 PL-259 N BNC F and SMA view 1.webp`, `Images/20-Type SMA Adapter Kit – Univer
- `data/labeled/embedder/2.4mm-F/video_0048_z70.jpg`, `data/labeled/embedder/2.4mm-F/video_0099_z70.jpg`
- `data/labeled/embedder/2.4mm-F/video_0039_z50.jpg`, `data/labeled/embedder/2.4mm-F/video_0090_z50.jpg`
```

```
- `data/labeled/embedder/2.4mm-F/synth_000050.jpg`, `data/labeled/embedder/2.4mm-F/synth_000109.jpg`
- `data/labeled/embedder/2.4mm-F/synth_000020.jpg`, `data/labeled/embedder/2.4mm-F/synth_000079.jpg`
- `data/labeled/embedder/2.4mm-F/synth_000001.jpg`, `data/labeled/embedder/2.4mm-F/synth_000060.jpg`
- `data/labeled/embedder/2.4mm-F/synth_000030.jpg`, `data/labeled/embedder/2.4mm-F/synth_000089.jpg`
- `data/labeled/embedder/2.4mm-F/video_0036_centralv2.jpg`, `data/labeled/embedder/2.4mm-F/video_0087_centralv2.jpg`
- `data/labeled/embedder/2.4mm-F/synth_000038.jpg`, `data/labeled/embedder/2.4mm-F/synth_000097.jpg`
- `data/labeled/embedder/2.4mm-F/video_0038_z50.jpg`, `data/labeled/embedder/2.4mm-F/video_0089_z50.jpg`
- `data/labeled/embedder/2.4mm-F/video_0033_z50.jpg`, `data/labeled/embedder/2.4mm-F/video_0084_z50.jpg`
- `data/labeled/embedder/2.4mm-F/synth_000016.jpg`, `data/labeled/embedder/2.4mm-F/synth_000075.jpg`
- `data/labeled/embedder/2.4mm-F/video_0042_z70.jpg`, `data/labeled/embedder/2.4mm-F/video_0093_z70.jpg`
- `data/labeled/embedder/2.4mm-F/video_0030_z70.jpg`, `data/labeled/embedder/2.4mm-F/video_0081_z70.jpg`
- `data/labeled/embedder/2.4mm-F/synth_000026.jpg`, `data/labeled/embedder/2.4mm-F/synth_000085.jpg`
- `data/labeled/embedder/2.4mm-F/video_0033_z70.jpg`, `data/labeled/embedder/2.4mm-F/video_0084_z70.jpg`
- `data/labeled/embedder/2.4mm-F/video_0048_central.jpg`, `data/labeled/embedder/2.4mm-F/video_0099_central.jpg`
- `data/labeled/embedder/2.4mm-F/synth_000013.jpg`, `data/labeled/embedder/2.4mm-F/synth_000072.jpg`
- `data/labeled/embedder/2.4mm-F/synth_000007.jpg`, `data/labeled/embedder/2.4mm-F/synth_000066.jpg`
- `data/labeled/embedder/2.4mm-F/video_0027_central.jpg`, `data/labeled/embedder/2.4mm-F/video_0078_central.jpg`
+ python -u -m rfconnectorai.data.crop_instances --input training/data/labeled/embedder --manifest datasets/rfconnectors/instances.jsonl --ou
whole-image: scanned 13850 images under training/data/labeled/embedder, skipped 0 unreadable, kept 13850
manifest: datasets/rfconnectors/instances.jsonl (13850 rows)
crops: datasets/rfconnectors/crops (0 files written)
+ wc -l datasets/rfconnectors/instances.jsonl
13850 datasets/rfconnectors/instances.jsonl
```

```
%%shell
set -euxo pipefail
cd /content/hotdogornot
export PYTHONPATH=/content/hotdogornot/training
export PYTHONUNBUFFERED=1

python -u -m rfconnectorai.data.build_yolo_dataset \
    --input datasets/rfconnectors/instances.jsonl \
    --out datasets/rfconnectors --base-dir training \
    --taxonomy training/rfconnectorai/specs/connectors.yaml

ls datasets/rfconnectors/
cat datasets/rfconnectors/data.yaml
cat datasets/rfconnectors/dataset.lock.json
find datasets/rfconnectors/images -name "*.jpg" | wc -l
```

```
+ cd /content/hotdogornot
+ export PYTHONPATH=/content/hotdogornot/training
+ PYTHONPATH=/content/hotdogornot/training
+ export PYTHONUNBUFFERED=1
+ PYTHONUNBUFFERED=1
+ python -u -m rfconnectorai.data.build_yolo_dataset --input datasets/rfconnectors/instances.jsonl --out datasets/rfconnectors --base-dir tra
{
  "created_at": "2026-05-12T07:00:19.981340+00:00",
  "dataset_id": "rfconnectors_2026_05_12_001",
  "dry_run": false,
  "family_to_idx": {
```

```
    "2.4MM": 0,
    "2.92MM": 1,
    "3.5MM": 2
  },
  "holdout_excluded": true,
  "instances_sha256": "5462b32c138b72681502bd3ac63c68fb394faace4ecc2ca143d6c962f58e8ee2",
  "skipped_count": 0,
  "split_seed": 1337,
  "taxonomy_sha256": "c87ddaf7b0e9bbe56a7a2efee62f08744b383ccf7f30575d3286cfde4a85ece7",
  "test_count": 1322,
  "train_count": 11117,
  "val_count": 1411
}
```


```
+ ls datasets/rfconnectors/
attributes.csv  dataset.lock.json  data.yaml  images  instances.jsonl  labels
+ cat datasets/rfconnectors/data.yaml
```


```
path: /content/hotdogornot/datasets/rfconnectors
train: /content/hotdogornot/datasets/rfconnectors/images/train
val: /content/hotdogornot/datasets/rfconnectors/images/val
test: /content/hotdogornot/datasets/rfconnectors/images/test
nc: 3
names:
  - 2.4MM
  - 2.92MM
  - 3.5MM
```


```
+ cat datasets/rfconnectors/dataset.lock.json
```


```
{
  "created_at": "2026-05-12T07:00:19.981340+00:00",
  "dataset_id": "rfconnectors_2026_05_12_001",
  "dry_run": false,
  "family_to_idx": {
    "2.4MM": 0,
    "2.92MM": 1,
    "3.5MM": 2
  },
  "holdout_excluded": true,
  "instances_sha256": "5462b32c138b72681502bd3ac63c68fb394faace4ecc2ca143d6c962f58e8ee2",
  "skipped_count": 0,
  "split_seed": 1337,
  "taxonomy_sha256": "c87ddaf7b0e9bbe56a7a2efee62f08744b383ccf7f30575d3286cfde4a85ece7",
  "test_count": 1322,
  "train_count": 11117,
  "val_count": 1411
}+ find datasets/rfconnectors/images -name '*.jpg'
```


```
+ wc -l
13849
```

```
%%bash
set -euxo pipefail
cd /content/hotdogornot
git pull --ff-only
```

```
export PYTHONPATH=/content/hotdogornot/training

# Re-build data.yaml with absolute paths
cd /content/hotdogornot/training
python -m rfconnectorai.data.build_yolo_dataset \
  --input ../datasets/rfconnectors/instances.jsonl \
  --out ../datasets/rfconnectors \
  --base-dir . \
  --taxonomy rfconnectorai/specs/connectors.yaml
cd /content/hotdogornot

# Verify the fix — should show absolute paths
cat datasets/rfconnectors/data.yaml

# Now retry detector
python -u -m rfconnectorai.detector.train_yolo \
    --data datasets/rfconnectors/data.yaml --model yolo11n.pt \
    --epochs 10 --imgsz 640 --batch 96 --device 0 \
    --dataset-lock datasets/rfconnectors/dataset.lock.json \
    --out reports/experiments/detector_run_001 \
    --artifact-out models/detector
```

```
Already up to date.
```


```
{
  "created_at": "2026-05-12T04:44:56.507883+00:00",
  "dataset_id": "rfconnectors_2026_05_12_001",
  "dry_run": false,
  "family_to_idx": {
    "2.4MM": 0,
    "2.92MM": 1,
    "3.5MM": 2
  },
  "holdout_excluded": true,
  "instances_sha256": "5462b32c138b72681502bd3ac63c68fb394faace4ecc2ca143d6c962f58e8ee2",
  "skipped_count": 0,
  "split_seed": 1337,
  "taxonomy_sha256": "c87ddaf7b0e9bbe56a7a2efee62f08744b383ccf7f30575d3286cfde4a85ece7",
  "test_count": 1322,
  "train_count": 11117,
  "val_count": 1411
}
path: /content/hotdogornot/datasets/rfconnectors
train: /content/hotdogornot/datasets/rfconnectors/images/train
val: /content/hotdogornot/datasets/rfconnectors/images/val
test: /content/hotdogornot/datasets/rfconnectors/images/test
nc: 3
names:
  - 2.4MM
  - 2.92MM
  - 3.5MM
```

```
Ultralytics 8.4.48 🚀 Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugment, batch=96, bgr=0.0, box=7.5, cache=False, cf
Overriding model.yaml nc=80 with nc=3

                   from  n    params  module                                       arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv             [3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv             [16, 32, 3, 2]
  2                  -1  1      6640  ultralytics.nn.modules.block.C3k2            [32, 64, 1, False, 0.25]
  3                  -1  1     36992  ultralytics.nn.modules.conv.Conv             [64, 64, 3, 2]
  4                  -1  1     26080  ultralytics.nn.modules.block.C3k2            [64, 128, 1, False, 0.25]
  5                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
  6                  -1  1     87040  ultralytics.nn.modules.block.C3k2            [128, 128, 1, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv             [128, 256, 3, 2]
  8                  -1  1    346112  ultralytics.nn.modules.block.C3k2            [256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF            [256, 256, 5]
 10                  -1  1    249728  ultralytics.nn.modules.block.C2PSA           [256, 256, 1]
 11                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 12             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 13                  -1  1    111296  ultralytics.nn.modules.block.C3k2            [384, 128, 1, False]
 14                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 15             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 16                  -1  1     32096  ultralytics.nn.modules.block.C3k2            [256, 64, 1, False]
 17                  -1  1     36992  ultralytics.nn.modules.conv.Conv             [64, 64, 3, 2]
 18            [-1, 13]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 19                  -1  1     86720  ultralytics.nn.modules.block.C3k2            [192, 128, 1, False]
 20                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
 21            [-1, 10]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 22                  -1  1    378880  ultralytics.nn.modules.block.C3k2            [384, 256, 1, True]
 23        [16, 19, 22]  1    431257  ultralytics.nn.modules.head.Detect           [3, 16, None, [64, 128, 256]]
```

```
import torch, gc
gc.collect()
torch.cuda.empty_cache()
print(f"GPU free: {torch.cuda.mem_get_info()[0]/1e9:.1f} / {torch.cuda.mem_get_info()[1]/1e9:.1f} GB")
```

```
GPU free: 15.5 / 15.6 GB
```

```
%%shell
set -euxo pipefail
cd /content/hotdogornot
git pull --ff-only
cd training && pip install -e '.[dev]' -q && cd ..
export PYTHONPATH=/content/hotdogornot/training
export PYTHONUNBUFFERED=1

# Verify attributes.csv has gender labels
echo '--- attributes.csv preview ---'
wc -l datasets/rfconnectors/attributes.csv
head -3 datasets/rfconnectors/attributes.csv
```

```
# 5-epoch smoke test with EfficientNetV2-S
python -u -m rfconnectorai.classifier.train_multihead \
    --dataset datasets/rfconnectors \
    --backbone efficientnet_v2_s \
    --epochs 10 --batch 32 --imgsz 384 --device 0 \
    --out reports/experiments/multihead_effnet_smoke \
    --artifact-out models/multihead_classifier \
    --base-dir training
```

```
+ cd /content/hotdogornot
+ git pull --ff-only
Already up to date.
+ cd training
+ pip install -e '.[dev]' -q
  Installing build dependencies ... done
  Checking if build backend supports build_editable ... done
  Getting requirements to build editable ... done
  Installing backend dependencies ... done
  Preparing editable metadata (pyproject.toml) ... done
  Building editable for rfconnectorai (pyproject.toml) ... done
+ cd ..
+ export PYTHONPATH=/content/hotdogornot/training
+ PYTHONPATH=/content/hotdogornot/training
+ export PYTHONUNBUFFERED=1
+ PYTHONUNBUFFERED=1
+ echo '--- attributes.csv preview ---'
--- attributes.csv preview ---
+ wc -l datasets/rfconnectors/attributes.csv
13851 datasets/rfconnectors/attributes.csv
+ head -3 datasets/rfconnectors/attributes.csv
instance_id,split,source_image,family,precision_family,side_a_gender,side_b_gender,polarity,mount_style,orientation,termination,finish_material
inst_0baa9288040720a2,train,data/labeled/embedder/2.4mm-F/video_0099_z70.jpg,2.4MM,not_applicable,female_socket,not_applicable,not_applicable,u
inst_26f1666ffd2a66e7,train,data/labeled/embedder/2.4mm-F/synth_000202.jpg,2.4MM,not_applicable,female_socket,not_applicable,not_applicable,unk
+ python -u -m rfconnectorai.classifier.train_multihead --dataset datasets/rfconnectors --backbone efficientnet_v2_s --epochs 10 --batch 32 --i
epoch  1/10 train_loss=0.332 val_loss=0.290 mean_val_acc=0.978
epoch  2/10 train_loss=0.286 val_loss=0.281 mean_val_acc=0.980
epoch  3/10 train_loss=0.280 val_loss=0.279 mean_val_acc=0.982
epoch  4/10 train_loss=0.276 val_loss=0.276 mean_val_acc=0.983
epoch  5/10 train_loss=0.272 val_loss=0.277 mean_val_acc=0.982
epoch  6/10 train_loss=0.268 val_loss=0.272 mean_val_acc=0.984
epoch  7/10 train_loss=0.263 val_loss=0.271 mean_val_acc=0.985
epoch  8/10 train_loss=0.260 val_loss=0.270 mean_val_acc=0.986
epoch  9/10 train_loss=0.257 val_loss=0.269 mean_val_acc=0.987
epoch 10/10 train_loss=0.257 val_loss=0.269 mean_val_acc=0.987
best checkpoint: reports/experiments/multihead_effnet_smoke/best.pt
predictions:     reports/experiments/multihead_effnet_smoke/predictions.jsonl (1322 rows)
metrics:         reports/experiments/multihead_effnet_smoke/metrics.json
```

```
!zip -r /content/full_pipeline.zip \
    /content/hotdogornot/reports/experiments/detector_run_001/ \
    /content/hotdogornot/models/detector/best.pt \
    /content/hotdogornot/reports/experiments/multihead_effnet_smoke/ \
    /content/hotdogornot/models/multihead_classifier/

from google.colab import files
files.download('/content/full_pipeline.zip')
```

```
    zip warning: name not matched: /content/hotdogornot/reports/experiments/detector_run_001/
  adding: content/hotdogornot/models/detector/best.pt (deflated 10%)
  adding: content/hotdogornot/reports/experiments/multihead_effnet_smoke/ (stored 0%)
  adding: content/hotdogornot/reports/experiments/multihead_effnet_smoke/head_vocabs.json (deflated 70%)
  adding: content/hotdogornot/reports/experiments/multihead_effnet_smoke/predictions.jsonl (deflated 89%)
  adding: content/hotdogornot/reports/experiments/multihead_effnet_smoke/best.pt (deflated 7%)
  adding: content/hotdogornot/reports/experiments/multihead_effnet_smoke/model_record.json (deflated 49%)
  adding: content/hotdogornot/reports/experiments/multihead_effnet_smoke/config.json (deflated 32%)
  adding: content/hotdogornot/reports/experiments/multihead_effnet_smoke/metrics.json (deflated 87%)
  adding: content/hotdogornot/models/multihead_classifier/ (stored 0%)
  adding: content/hotdogornot/models/multihead_classifier/best.pt (deflated 7%)
```